HAFIZ MUHAMMAD

AI Engineer • Agentic AI & LLMs • RAG Pipelines • Python • FastAPI
Lahore, Pakistan | 0345-0976304 | hafizmuhammadshah11@gmail.com
LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

AI Engineer with 1+ year of hands-on experience designing and deploying Agentic AI systems and LLM-driven applications. Specialist in building Agentic Workflows using LangChain and LangGraph, implementing RAG architectures for intelligent data retrieval, and integrating AI models (OpenAI-compatible APIs, LLaMA3, Mistral-style open-source models) into production systems. Experienced in Voice-to-Text (Whisper AI), Text-to-Speech (Cartesia TTS), and Computer Vision (OpenCV, Wav2Lip) pipelines. Currently building production AI backends at PureLogics using Python and FastAPI. Available to start immediately and comfortable working onsite in Lahore.

TECHNICAL SKILLS

Agentic AI & LLM Frameworks

LangChain & LangGraph: Primary frameworks for building Agentic Workflows, multi-step reasoning, tool use, and autonomous agent design

LLM Integration: OpenAI-compatible APIs (Groq, Ollama), LLaMA3, Mistral-style open-source models — fine-tuning and prompt engineering

Prompt Engineering: Advanced prompt design, chain-of-thought reasoning, few-shot prompting, and system instruction optimization

RAG Architectures: Full RAG pipeline design — embedding generation, vector indexing, hybrid BM25 + semantic retrieval for real-time data processing

AI Libraries & Vector Databases

PyTorch: Model integration and inference pipelines for deep learning and AI application development

ChromaDB: Vector database for semantic search and embedding storage — production RAG implementations

allMiniLM-L6-v2: Sentence embedding model for dense vector generation in retrieval pipelines

Pinecone / Weaviate: Familiar with vector database architecture; able to adapt from ChromaDB to Pinecone or Weaviate quickly

Multimodal AI & Voice

Whisper AI: Voice-to-Text — speech transcription pipelines with 95% accuracy for meeting recordings and real-time audio

Cartesia TTS / Coqui TTS: Text-to-Speech — low-latency voice synthesis integrated into conversational AI applications

OpenCV & Wav2Lip: Computer Vision — spatial processing, image analysis, and AI-driven avatar lip-sync for multimodal applications

PyannoteAI: Speaker diarization for multi-speaker audio processing and structured transcript generation

Backend & Infrastructure

Python: Advanced — primary language for all AI development, backend microservices, and automation pipelines

FastAPI: High-performance async Python framework for building backend microservices supporting high-concurrency AI requests

Flask & Django: Python web frameworks for REST API development and server-side AI application delivery

MySQL & SQLAlchemy: Relational database management, ORM-based data access, and schema optimization

Git & Linux: Version control, Linux server management, and production deployment practices

PROFESSIONAL EXPERIENCE

AI Engineer | **PureLogics, Lahore** | *July 2025 – Present*

- Design and deploy Agentic AI workflows using LangGraph and LangChain — building autonomous agents with multi-step reasoning, tool use, and real-time LLM API integration.
- Implement RAG architectures for document search and knowledge retrieval using ChromaDB vector database and allMiniLM-L6-v2 embeddings integrated with LLM backends.
- Build backend microservices in Python with FastAPI to support high-concurrency AI requests; integrate OpenAI-compatible APIs (Groq, Ollama) into production-grade applications.
- Develop multimodal AI pipelines combining Whisper AI (Voice-to-Text), TTS systems, and computer vision (OpenCV) for spatial and audio-driven AI interactions.

PORTFOLIO & PROJECTS

All projects with full GitHub source code: **hafizmuhammadshah42895.github.io/hafizmuhammad.dev**

Movers – Agentic AI Logistics Platform — *Lead AI Engineer | 2025*

- Designed and deployed an Agentic Workflow using LangGraph — autonomous dispatch agent with multi-step reasoning, real-time tool use, and LLM-driven decision-making.
- Implemented a full RAG architecture (ChromaDB + allMiniLM embeddings) for real-time logistics knowledge retrieval; integrated Whisper AI (Voice-to-Text) and Cartesia TTS for voice coordination.

Stack: Python, FastAPI, LangGraph, LangChain, RAG, ChromaDB, Groq (OpenAI API), Whisper AI, Cartesia TTS, MySQL

Gourmet Haven – Conversational AI Voice Agent — *AI Engineer | 2025*

- Built an Agentic conversational AI system with LangGraph state management and hybrid BM25 + Vector Search RAG pipeline for grounded, real-time factual responses.
- Integrated OpenAI-compatible Groq LLM with Cartesia TTS achieving sub-second voice latency — demonstrating Text-to-Speech pipeline optimization for production.

Stack: Python, LangGraph, LangChain, RAG, ChromaDB, BM25, Groq LLM, Ollama, Cartesia TTS, Flask

AI-Powered Video Assistant — *ML & AI Developer | 2025*

- Built a multimodal AI pipeline: Whisper AI (Voice-to-Text, 95% accuracy), PyannoteAI (speaker diarization), ChromaDB semantic indexing, and Ollama LLM for spatial video Q&A.
- Designed the full embedding pipeline from raw audio/video ingestion to vector storage and conversational retrieval — a real-time spatial data processing system.

Stack: Python, Whisper AI, PyannoteAI, ChromaDB, Ollama LLM, Embedding Pipeline, Flask, FFmpeg

Multimodal AI Document Chat — *AI Solutions Developer | 2025*

- Built a multimodal RAG system using LLaMA3, ChromaDB vector store, Wav2Lip (Computer Vision for spatial interaction), and Coqui TTS — all unified in one AI application.
- Integrated OpenCV for visual processing and FFmpeg for audio-video synchronization — demonstrating Computer Vision capabilities for spatial AI applications.

Stack: Python, LLaMA3, ChromaDB, RAG, Wav2Lip, OpenCV, Coqui TTS, Whisper AI, Flask

EDUCATION & CERTIFICATIONS

B.Sc. Computer Science — Riphah International University, Lahore | Graduated Jan 2026

Certifications: Databases & SQL for Data Science (IBM) • OOP in Java (Oracle) • Design Patterns (FreeCodeCamp) • Linux Management (Linux Foundation)